

Advanced Lab - Experiment 5: Nuclear Fission

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1 Abstract

Heavy nuclei at the earth's surface have been undergoing spontaneous fission since the beginning of their creation. This experiment investigates the principles and behavior of nuclear fission, with a focus on the simulation of chain reactions using the Monte Carlo method. Fission occurs when a heavy nucleus, such as uranium-235, absorbs a neutron and splits into two smaller nuclei, releasing a large amount of energy and additional neutrons. These emitted neutrons can induce further fissions, potentially leading to a self-sustaining chain reaction. The lab explores conditions necessary for such a reaction, including the concepts of critical mass, neutron moderation, and the multiplication factor (k).

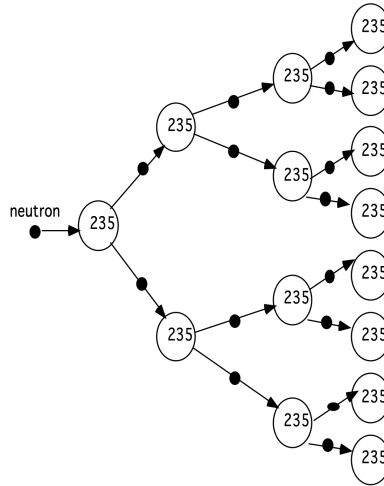


Figure 1: Visual of fission chain reaction

2 Introduction

To help us better understand how these chain reactions work we created a simulation in python to find an experimental multiplication factor. This simulation aims to model nuclear fission using a Monte Carlo simulation to explore how factors like neutron absorption, escape probability, and material geometry affect the likelihood of a sustained chain reaction. Each instance of fission releases 2 neutrons which each have a change of induced fission, creating a chain reaction. By calculating the survival fraction of neutrons capable of producing further fissions, we estimate the conditions required for a system to reach the critical value. The critical value is reached when each fission on average produces one induced fission. This computational approach provides insight into the stochastic nature of neutron behavior and the delicate balance required to safely harness nuclear energy.

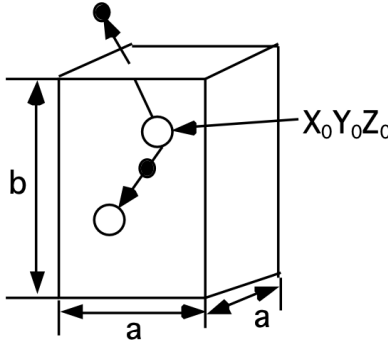


Figure 2: Two neutrons being emitted during fission

The image above shows an example of fission. Two Neutrons are emitted, one inside the box and one outside the box. The one outside the box will not create induced fission, but the one inside will, starting the chain reaction. Our simulation will simulate this exact situation over and over again testing to see how many chain reactions will happen based on shapes of different mass and size.

3 Procedure

The first step in this lab was to create a functional simulation that would find us an f value based on mass (M) of a uranium block, the ratio of the 2 sides of the block (S) and N the number of random fissions. To create this simulation I used 4 different functions in python. The first computed block dimensions (a, b) based on the given M and S values. The next simulated random fission. Using a system of random variables we simulated random starting coordinates, X_0, Y_0, Z_0 and random endpoint coordinates, $X_1, Y_1, Z_1, X_2, Y_2, Z_2$. There are

2 endpoints because 2 neutrons are emitted from each fission. Once we had this data I created another function to test if the new endpoints were inside of the Uranium block. If they were we added one to the "hits" variable, a successful reaction. The final function looped through this random fission function based on the N, number of fissions specified originally.

Now that we had a working simulation it was time to use it to find some data. First we wanted to find the critical mass when the shape of the box was a cube, or S was equal to 1 and all the side lengths were equal. To do this we slowly increased the value of M starting at 0 keeping N and S constant. Once we reached $f = 1$ or around there we knew we were about at the critical mass. A value of $f = 1$ meant that on average for each fission there would be one induced fission, causing a chain reaction. but not an explosion. This is of course a very important number.

Secondly we wanted to find the critical mass when S was more then or greater then one. To do this we ran a similar test but for the S values .5 and 1.5 still keeping N constant.

Finally we wanted to simulate this fission inside of a sphere instead of a cube or rectangle. To do this we set the radius to $a/2$ and kept a similar code for the rest. Using this code we found the critical value of M.

4 Data

M	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1	1.1	1.2	1.3	1.4	1.5	1.6
F	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
	0.528	0.627	0.651	0.743	0.74	0.807	0.824	0.88	0.851	0.891	0.944	0.973	0.942	0.994	1.001
	0.515	0.587	0.678	0.684	0.72	0.796	0.812	0.853	0.874	0.947	0.969	0.938	1.018	0.994	1.002
	0.514	0.576	0.683	0.723	0.705	0.812	0.814	0.891	0.892	0.863	0.941	0.961	0.995	1.027	0.996
	0.503	0.588	0.679	0.727	0.735	0.778	0.85	0.816	0.882	0.848	0.904	0.969	0.96	0.998	1.045
	0.503	0.606	0.652	0.692	0.762	0.816	0.835	0.842	0.841	0.907	0.946	0.957	0.995	0.987	0.97
Average F	0.5126	0.5968	0.6686	0.7138	0.7324	0.7998	0.827	0.8564	0.868	0.8912	0.9408	0.9596	0.982	1	1.0028

Figure 3: Critical Value Data for S=1

M	1.3	1.35	1.32	1.31
1	0.947	1.006	0.967	1.01
2	0.977	1.025	1.022	0.981
3	1.015	1.055	1.03	0.982
4	1.029	0.982	0.993	1.005
5	1.016	0.982	1.004	1.033
Average F	0.9968	1.01	1.0032	1.0022

Figure 4: Critical Value Date for S=.5

M	5.5	5.6	5.7	5.8	5.9	6	6.1
1	0.955	0.975	0.977	0.943	0.982	0.996	0.999
2	0.998	0.96	0.922	0.97	0.975	1.024	1.038
3	0.99	1.019	1.037	0.982	0.986	0.973	0.942
4	0.982	0.997	0.993	0.939	0.946	0.954	1.01
5	0.999	0.992	1.031	0.979	0.971	1.053	1.012
Average F	0.9848	0.9886	0.992	0.9626	0.972	1	1.0002

Figure 5: Critical Value Date for S=1.5

M	1.3	1.4	1.45	1.43
1	0.969	0.989	1.027	0.979
2	1.014	1.023	1.025	0.996
3	0.968	0.981	0.956	1.002
4	0.997	0.986	1.04	1.003
5	0.961	0.978	1.048	1.035
Average F	0.9818	0.9914	1.0192	1.003

Figure 6: Critical Value for Sphere

5 Results

Looking at our simulations it looks like for the rectangle of uranium when $S = 1$, the critical value of M was about 1.5, when $S = .5$, the critical value of M was about 1.31 and for $S = 1.5$ the critical value for M was about 6. Judging from these results we can assume that the relationship between S and the critical value of M is positively exponential and as S grows the critical value grows very quickly. Looking at the sphere we got a critical value of approximately 1.43. These values align and are consistent with theoretical expectations from previous studies.

6 Conclusion

In this experiment, we successfully used a Monte Carlo simulation to model nuclear fission chain reactions and determine the critical mass for uranium blocks of varying geometries. Our results demonstrate that the shape and size of the fissile material significantly impact the multiplication factor f and thus the conditions required for the critical value. For rectangular blocks, as the side ratio S increased, the critical mass M grew rapidly, suggesting a strong geometric influence on neutron escape probabilities and reaction sustainability. The spherical geometry, known for its minimal surface area to volume ratio, showed a slightly

lower critical mass compared to elongated shapes, confirming theoretical expectations about neutron containment. Overall, this simulation shows the delicate balance of parameters needed to sustain a chain reaction safely, highlighting the importance of geometry in nuclear reactor design and criticality safety.